

Membrane Air Separation

Group #2
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1

Introduction

- **Purpose of lab**
 - The purpose of this lab is to find the transport coefficient for oxygen and nitrogen using the relationships of Henry's Law and Fick's Law.

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2

Theory

- The components ability to diffuse is based on Fick's Law seen in equation (1) below:

$$J_A = D_A \left(\frac{C_{A1} - C_{A2}}{z} \right) \quad (1)$$

- The total flux of component A is calculated using equation (2) seen below:

$$J_A = \frac{V_{sep} \rho}{nA} \quad (2)$$

Where:
 J_A = flux of component A in $\frac{mol}{m^2 \cdot s}$
 D_A = diffusivity of component A in $\frac{m^2}{s}$
 z = thickness of membrane in meters
 C_{A1} = concentration of component A inside membrane wall on feed side in $\frac{mol}{m^3}$
 C_{A2} = concentration of component A outside membrane wall on permeate side in $\frac{mol}{m^3}$

Where:
 V_{sep} = volumetric flow rate of components A in the permeate in $\frac{m^3}{s}$
 ρ = density of permeate in $\frac{mol}{m^3}$
 A = area of membrane in m^2
 n = number of modules used

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3

Theory Continued...

- The components ability to dissolve is governed by Henry's Law seen in equation (3) below:

$$C_A = p_A S_A \quad (3)$$

Where:
 C_A = concentration of component A inside membrane wall in $\frac{mol}{m^3}$
 S_A = solubility constant for component A in the membrane in $\frac{mol}{m^3 \cdot Pa}$
 p_A = partial pressure of component A in the gas phase in Pa

- Equation (4) below represents how the permeability coefficient is calculated:

$$q_A = D_A p_A S_A \quad (4)$$

Where:
 q_A = permeability coefficient in Barrer
 D_A = diffusivity of component A in $\frac{m^2}{s}$
 S_A = solubility constant for component A in the membrane in $\frac{mol}{m^3 \cdot Pa}$

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4

Slide 1

DL1 99/100
Donglin Li, 10/29/2019

Slide 3

DL2 Good for typing the equation. The font is too small and you can divide this slide into 2.
Donglin Li, 10/29/2019

DL3 -1
Donglin Li, 10/29/2019

Theory Continued ...

- Now equations (1), (3), and (4) are combined to give a simpler calculation seen in equation (5):
 - $J_A = q_A(P_{A1} - P_{A2})$ (5)

Where:
 q_A = permeability of component A
 P_{A1} = partial pressure of component A on the feed side
 P_{A2} = partial pressure of component A on the permeate side

- Calculating the overall mass transfer coefficients involves equation (6) and (7):
 - $O_2 \text{ flow rate} = K_{O_2}(\text{O}_2 \text{ tube pressure} - \text{O}_2 \text{ shell pressure})$ (6)
 - $N_2 \text{ flow rate} = K_{N_2}(\text{N}_2 \text{ tube pressure} - \text{N}_2 \text{ shell pressure})$ (7)

Theory Continued...

- To calculate the efficiency of the separation equation (8) is implemented. (α_{AB} represents the efficiency)

- $\alpha_{AB} = \frac{y_A}{x_B}$ (8)

- In this experiment the efficiency was calculated using equations (9) and (10)

- $\alpha'_{AB} = \left(\frac{y_{AB}}{x_{AB}} \right) \left(\frac{x_{AF}}{y_{AF}} \right)$ (9)

- $\alpha''_{AB} = \left(\frac{x_{AB}}{x_{AF}} \right) \left(\frac{y_{BF}}{y_{BF}} \right)$ (10)

Where:
 α'_{AB} = separation factor based on retentate composition
 x_{AB} = mole fraction of component A in permeate
 x_{AF} = mole fraction of component A in feed
 y_{AB} = mole fraction of component B in permeate
 y_{BF} = mole fraction of component B in feed
 α''_{AB} = separation factor based on feed composition
 x_{AB} = mole fraction of component A in permeate
 x_{AF} = mole fraction of component A in feed
 y_{BF} = mole fraction of component B in feed

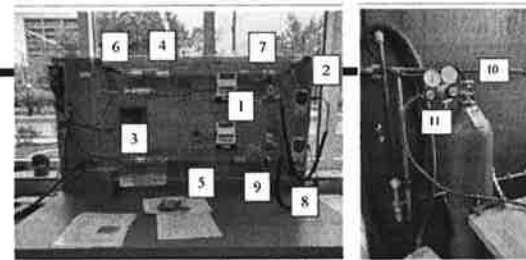
Theory Continued...

- The recovery of oxygen and nitrogen are defined by equations (11) and (12) seen below:

- $O_2 \text{ recovery} = \frac{V_p C_{O_2p}}{V_f C_{O_2f}}$ (11)

- $N_2 \text{ recovery} = \frac{V_p C_{N_2p}}{V_f C_{N_2f}}$ (12)

Where:
 V_p = volumetric flowrate of permeate in $\frac{m^3}{s}$
 V_f = volumetric flowrate of feed in $\frac{m^3}{s}$
 C_{O_2p} = molar concentration of component A in permeate in $\frac{mol}{m^3}$
 C_{O_2f} = molar concentration of component A in feed in $\frac{mol}{m^3}$



Apparatus

1. Prism Module - non-porous semipermeable membranes
2. Oxygen Analyzer - measures O_2 content
3. Digital Pressure Reader - air flow through pipes
4. Shell - top cylinder
5. Tube - bottom cylinder
- 6 - 9. Valves A-D - Used to purge or change configuration
10. Pressure Valve - Allow air to enter system
11. Pressure Regulator - Control air flow

Material & Supplies

- Air is the only material needed for this experiment.

Procedure

- Turn on the pressure cylinder to 140 psig
- Turn on the oxygen analyzers and allow 10 minutes for it to reach a steady reading
- Turn the power on and set up the valves in parallel configuration
 - Valve A- Left
 - Valve B- Up
 - Valve C- Down
 - Valve D- Up

Procedure Continued...

- Turn the pressure to the wanted operating pressure and the wanted flowrate
- Wait a few minutes to get a steady reading and record the tube and shell oxygen levels as well as the flowrate for the shell
- After all the runs are finished, change the valves to series configuration by changing
 - Valve A- Right
 - Valve C- Up

Procedure Continued...

- Repeat the same procedure and complete each of the runs

Experimental Challenges

- Make sure to turn the valves B and D up and not down to the vent, this doesn't give you the correct reading
- Make sure to watch the pressure as it can fluctuate when changing the flowrates
- Watch the flowrates as it fluctuates as well

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13

Safety

- Make sure not to put a lot of pressure, keep under 200 psig
- Make sure the needle doesn't change on the valves

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14